Career Description

Research Intern Application

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Research Profile

Computer vision researcher focused on **image restoration, computational photography, and data augmentation**. My research centers on developing physically grounded and data-driven vision models for robust image restoration under real-world imaging conditions. Experience spans differentiable image formation, industrial display imaging, large-scale deep-learning experimentation, and real-time inference optimization.

Research Focus

Image Restoration · Computational Photography · Data Augmentation · Computer Vision

Technical Strengths

PyTorch · CUDA · Distributed Training · TensorRT · Docker · Linux

Selected Publications

2027	**Trajectory-aware Differentiable Blur Augmentation for Image Deblurring** Target Venue: ICLR 2027, ongoing research
2026	**DnSAD: Defocus and Spatially Adaptive Deblurring for Mura Compensation in Display Panels** Journal of the Society for Information Display, Co-first Author
2025	**Intrinsic Decomposition-based Low-Light Image Enhancement** IEEE Signal Processing Letters

Selected Projects

2024	**Samsung AI Challenge — Machine Learning Force Fields** 2nd Place, Energy/Force Prediction and OOD Uncertainty Estimation
2023	**Vision-based Safety Equipment Inspection** KIST AI·Robot Institute, real-time vision pipeline and TensorRT deployment

This document summarizes the technical scope, individual contributions, experimental methodology, and measurable outcomes of selected research and engineering projects.

Trajectory-aware Differentiable Blur Augmentation for Image Deblurring

Period	2026 – Present	**Context**	Korea University, Deep Image Processing Lab
Role	Primary researcher	**Tech**	PyTorch, CUDA, DDP, Mixed Precision, LMDB

Goal. Improve deblurring generalization by moving beyond conventional blur augmentation that primarily varies motion magnitude and orientation. The project explicitly models **diverse motion trajectories** and introduces a differentiable augmentation process whose motion perturbations can be optimized with respect to the deblurring objective.

Problem & Motivation

Motion blur is caused by camera shake and object motion during exposure, and the temporal evolution of the underlying motion trajectory strongly affects the resulting blur pattern. Existing blur augmentation methods mainly manipulate coarse motion attributes, which can underrepresent the diversity of real camera and object trajectories.

The research focuses on two questions:

- How can blur augmentation represent richer temporal and geometric motion trajectories?
- Can augmentation be learned to generate challenging and informative blur patterns instead of relying only on predefined random perturbations?

Approach

I designed a **trajectory-aware reblur and augmentation framework** with three core components:

- **Trajectory-aware reblur modeling.** A reblur network learns a compact trajectory representation from blur–sharp pairs and reconstructs the observed blur through differentiable image formation.
- **Global and local motion modeling.** Global camera motion and spatially varying local object motion are modeled jointly to represent realistic heterogeneous blur.
- **Learnable differentiable augmentation.** A motion augmentation network perturbs the learned trajectory parameters and is optimized through the deblurring objective to generate challenging yet controlled training samples.

Technical Challenges

- Designing a compact motion representation that captures acceleration, curved trajectories, local motion, and rotational dynamics while remaining stable for optimization.
- Preserving differentiability through trajectory construction, warping, and temporal blur synthesis.
- Balancing augmentation difficulty so that generated blur is challenging without becoming physically implausible.
- Performing controlled limited-data experiments across multiple dataset scales and deblurring backbones.

Contributions

- Designed the trajectory parameterization and global–local motion representation used by the reblur network.
- Developed the differentiable blur synthesis and learnable motion augmentation framework.
- Implemented the end-to-end training pipeline with mixed-precision and distributed training support.
- Designed limited-data experiments and conducted benchmark evaluations on GoPro and HIDE.

Current status. Ongoing research targeting ICLR 2027. The proposed method consistently improves deblurring performance on GoPro and HIDE, with the largest gains observed in limited-data training settings.

Trajectory-aware Differentiable Blur Augmentation — Results

+2.45 dB
GoPro 10% vs. baseline

31.37 dB
GoPro 10%, MIMO-UNet + Ours

28.98 dB
HIDE, 10% training setting

Quantitative Results

	GoPro 10%		GoPro 100%	
Model	PSNR	SSIM	PSNR	SSIM
MIMO-UNet+	28.92	0.915	32.44	0.957
+ ID-Blau	30.64	0.940	32.93	0.961
+ **Ours**	**31.37**	**0.947**	**33.10**	**0.962**

HIDE Benchmark

	GoPro 10% Training		GoPro 100% Training	
Model	PSNR	SSIM	PSNR	SSIM
MIMO-UNet+	26.71	0.877	30.00	0.930
+ ID-Blau	28.50	0.910	30.68	0.938
+ **Ours**	**28.98**	**0.919**	**30.71**	**0.938**

What the Results Show

- The largest gain appears in the **low-data regime**, improving GoPro 10% performance by **2.45 dB** over the MIMO-UNet+ baseline.
- The method also outperforms ID-Blau, suggesting that explicitly diversifying motion trajectories provides benefits beyond magnitude- and orientation-based augmentation.
- Improvements transfer to the HIDE benchmark, indicating stronger generalization beyond the training distribution.
- Performance also improves in the full-data setting, showing that the method remains effective beyond data-scarce scenarios.

Research takeaway. Trajectory-aware augmentation is particularly effective when training data are limited, where expanding motion-space coverage can compensate for reduced diversity in the observed training distribution.

Current Direction

Current work focuses on improving trajectory diversity, evaluating additional deblurring architectures, and extending experiments across broader training-set sizes and realistic blur benchmarks.

DnSAD: Defocus and Spatially Adaptive Deblurring

Period	2025 – 2026	**Context**	LG Display industry-academia project
Role	Co-first author / researcher	**Tech**	PyTorch, Wiener Deconvolution, PSF Modeling

Industrial Problem. Display mura inspection commonly uses defocused capture to suppress moiré artifacts, but defocus and optical distortion introduce **spatially varying blur**. DnSAD explicitly leverages measured PSF maps to restore fine mura structures while adapting the restoration process to spatially varying optical degradation.

Approach & Contributions

- Designed a **PSF-aware deblurring framework** that extends deep Wiener deconvolution from a single shift-invariant PSF to multiple spatially varying PSFs.
- Introduced **mask-guided spatial attention** to encode the spatial locations of measured PSFs and selectively fuse region-specific deblurred features.
- Added a **coarse-to-fine multi-scale refinement** module to reconstruct fine mura structures and high-frequency details.
- Constructed and evaluated a real-world OLED panel dataset with spatially aligned PSF maps and accurate ground-truth images.

Dataset & Experimental Setup

The real-world dataset was captured from actual OLED panels using a high-resolution industrial camera. Spatially varying PSFs were measured from dot-pattern captures, while a shift-dot acquisition process reconstructed accurate full-resolution ground-truth images.

The final dataset contains training and validation patches from two panels and **16 full-resolution 3840×2160 images** from a separate panel for evaluation.

Results

38.17 dB
Synthetic Flickr2K PSNR

36.41 dB
Real-world OLED PSNR

0.6474
Lowest MAE-MURA

On the synthetic benchmark, DnSAD achieved **38.17 dB PSNR / 0.9743 SSIM**.

On the real-world OLED panel dataset, DnSAD achieved **36.41 dB PSNR / 0.8944 SSIM** and the **lowest MAE-MURA of 0.6474**, outperforming both blind and non-blind deblurring baselines.

Ablation Insight

The largest performance gain came from modeling multiple spatially varying PSFs. The DWDN baseline improved from **31.47 dB to 38.04 dB** after introducing Multi-PSF restoration. Spatial attention and multi-scale refinement further improved the result to **38.17 dB**.

Impact. This project integrates **optical system modeling and deep image restoration** for an industrial display-inspection problem. It demonstrates how measured physical priors can be incorporated into a deep-learning pipeline to satisfy practical mura-compensation requirements.

Machine Learning Force Fields — Samsung AI Challenge 2024

Period	2024	**Context**	Samsung AI Challenge
Role	Team researcher	**Tech**	MACE, GNN, GPR, Ensemble Learning

Goal. Develop a machine learning force field for accurate molecular energy and force prediction, together with an uncertainty estimation framework for identifying out-of-distribution samples.

My Contributions

- Trained and optimized a **MACE-based message passing neural network** for molecular energy and force prediction.
- Performed extensive hyperparameter tuning over optimization, loss weighting, graph construction, and model configuration.
- Explored architectural modifications including Product Layer, Readout Layer, residual connections, DropEdge, and dropout.
- Designed ensemble strategies using models with different configurations to improve generalization.
- Developed a **Gaussian Process Regression-based OOD uncertainty estimation framework** and evaluated multiple kernel functions.

Optimization Strategy

The project began with benchmarking diverse model families, including point-cloud, neural-potential, equivariant, and message-passing approaches. MACE showed the strongest initial performance and was selected as the primary model.

Performance was then improved through systematic hyperparameter exploration and weighted ensemble learning. For uncertainty estimation, multiple Gaussian Process kernels were evaluated, with the periodic kernel achieving the strongest OOD detection performance.

Results

5.56 → 2.66
EF Score

2nd Place
Public EF leaderboard

0.9994
OOD AUROC, 1st Place

The final system reduced the public EF Score from **5.56 to 2.66** and achieved **2nd place**. The Gaussian Process-based uncertainty estimation model achieved **0.9994 OOD AUROC**, ranking **1st on the public leaderboard**.

What this project demonstrates. Experience with graph neural networks, scientific machine learning, systematic architecture and hyperparameter exploration, ensemble learning, and probabilistic uncertainty estimation.

Vision-based Safety Equipment Inspection — KIST

Period	2023	**Context**	KIST AI·Robot Institute
Role	Research intern	**Tech**	PyTorch, YOLOv7, EfficientNet, TensorRT

Goal. Develop a real-time laboratory safety inspection system that detects a person, checks personal protective equipment (PPE), performs second-stage classification for ambiguous equipment classes, and supports laboratory access control.

My Contributions

- Built a **17K-image PPE dataset** and established a structured data collection pipeline covering diverse poses, camera angles, lighting conditions, and locations.
- Improved YOLOv7 detection performance through additional web-crawled data, data augmentation, and hyperparameter optimization.
- Developed a **multi-stage vision pipeline** combining human detection, PPE detection, and dedicated shoe/glasses classification.
- Developed EfficientNet-based second-stage classifiers for difficult small-object categories.
- Optimized the entire inference pipeline with TensorRT and integrated it into a real-time laboratory access-control kiosk.

Model & System Optimization

The initial PPE detector frequently confused visually similar categories, such as shoes versus sandals and glasses versus safety glasses. To address these errors, the detector output was cropped and passed to dedicated second-stage classification models.

Human detection was additionally introduced to isolate the primary subject, align the training and inference distributions, and increase the proportion of useful pixels in each input.

Results

92.2 → 98.0
Detection mAP

96.0 → 98.8
Shoe classification

71 ms → 19 ms
Total inference latency

Block	Original	TensorRT
Human Detection	7 ms	6 ms
PPE Detection	13 ms	9 ms
Classification	51 ms	4 ms
Total	**71 ms**	**19 ms**

Engineering Takeaway

The classification stage was the major inference bottleneck. TensorRT reduced classification latency from **51 ms to 4 ms**, bringing total inference time from **71 ms to 19 ms**.

The optimized models were integrated into a functional laboratory access-control kiosk, providing end-to-end experience from data collection and model training to inference optimization and deployment.

What this project demonstrates. End-to-end computer-vision engineering: dataset construction, error analysis, detection and classification, multi-stage system design, inference acceleration, and real-world deployment.